Toan Vo

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Professional Summary

Software Developer and Machine Learning enthusiast with 3 years of experience in software development and 3 years of hands-on experience in machine learning. Proficient in software engineering with strong expertise in Python, C/C++, and JavaScript. In-depth knowledge of modern software frameworks such as NodeJS and ReactJS. Experienced in developing machine learning models, with a focus on Deep Learning and Data Analysis. Skilled in popular ML frameworks such as scikit-learn, PyTorch, TensorFlow, and JAX, with a growing interest in foundational AI, ML interpretability and its applications in real-world problems.

Education

University of Maryland - College Park — College Park, MD
M.S., Computer Science — *August 2026 – Present*

University of Wisconsin - Madison — Madison, WI
B.S., Computer Science. Certificate: Statistics, Mathematics. GPA: 3.7/4 — *August 2022 – May 2026*

- **Relevant courses**: Algorithms, Data Structures, Statistical Modeling, Object-Oriented Programming, Operating Systems, Database Systems, Probability Theory, Linear Algebra, Optimization, Deep Learning, Interpretable ML

Research and Technical Product

BRO: Bayesian Retraction Optimization for Tissue Attachment Mapping in Surgical Dissection — June 2024 – November 2024

- Co-authored of a research paper to be submitted to RA-L
- Developed a Bayesian robotic tissue-dissection framework that learns attachment probabilities in real time using Sequential Bayesian Hilbert Maps and optimizes retraction actions for safer, more efficient autonomous surgery.
- Implemented Bayesian optimization acquisition functions (EI, UCB, PI) for autonomous tissue-retraction on the dVRK, conducting experiments, generating performance analyses, and delivering detailed technical reports.

Experience

Undergraduate Researcher (AI semantics) — September 2025 – Present
Knowledge and Concepts Lab, University of Wisconsin-Madison — *Madison, WI*

- Conducted research on interpretability and control in neural networks, analyzing how internal representations vary across task demands.
- Evaluated semantic representations using SVM probing, dimensionality reduction, pairwise distance analysis, and multidimensional scaling (MDS).

Software Engineer Intern — June 2025 – August 2025
Viettel Solutions — *Ho Chi Minh City, Vietnam*

- Designed and containerized AI model evaluation environments using Docker, standardizing workflows and ensuring reproducibility across platforms.
- Implemented local LLM inference pipelines via Ollama, enabling rapid model testing and benchmarking without dedicated GPU infrastructure.
- Presented technical findings to senior engineers, including projected productivity gains, resource requirements, and deployment recommendations.
- Optimized Viettel's AI Tessel computer vision pipeline for multi-angle grocery shelf image analysis, increasing matching accuracy by 30% and reducing end-to-end processing time by 25%.

Undergraduate Researcher (Machine Learning and Robotics) — May 2024 – August 2024
KuntzLab, University of Utah — *Salt Lake City, UT*

- Developed a novel LSTM-based model to predict tendon robot shapes from current configurations, surpassing prior state-of-the-art performance by 21%.
- Engineered a Bayesian optimization framework to automate surgical task execution, improving tissue retraction efficiency by 27% and attachment point detection accuracy by 15%.
- Contributed to lab publications and presentations, demonstrating applicability of ML techniques in robotic surgical automation.

Projects

Deep Learning for Ocular Disease Classification | *Python, PyTorch* March 2025 – April 2025

- Built deep learning models to classify 8 common ocular diseases from ODIR-5K retinal images (6,392 total), achieving 50% classification accuracy.
- Conducted systematic evaluation of ResNet, EfficientNet, and custom CNN architectures, selecting optimal networks based on validation performance.
- Integrated Grad-CAM for visual interpretability, enabling clinical validation of model predictions by highlighting optic disc and macula regions.

MiniSpark | *C* April 2025

- Built MiniSpark, a single-node implementation of Apache Spark in C, supporting RDD DAGs (Resilient Distributed Datasets) with transformations (map, filter, join, partitionBy) and actions (count, print).
- Implemented a thread pool scheduler with locks and condition variables to materialize RDD partitions in parallel, added metrics logging for profiling, and ensured safe memory management.

Workspace Manager | *MongoDB, Javascript* November 2023

- Developed an application to streamline workspace setup and enhance workflow efficiency for multiple users.
- Implemented MongoDB-based backend to manage settings and preferences, reducing setup time by 50%.
- Incorporated secure authentication and encrypted accounts, supporting unlimited user customization.

SAT Score Analysis and Attendance Impact | *R* October 2022 – December 2022

- Conducted statistical analysis to determine correlation between school attendance and SAT performance across New York high schools.
- Utilized R for data cleaning, processing, and hypothesis testing, identifying significant differences in reading and writing scores (mean difference: 6.54 points).

Highway Crossing 3D Game | *OpenGL, C#* January 2022 – May 2022

- Designed and implemented a 3D highway crossing simulation with audio effects and textured 3D models for immersive gameplay.
- Optimized collision detection and OpenGL rendering pipeline, reducing latency by 20% and improving responsiveness.

Technical Skills

- **Programming & Data Analysis**: Python (advanced, ML/AI modeling, deep learning), Java & C/C++ (algorithm design, robotics simulation), SQL & R (data management, statistical analysis).
- **Tools & Platforms**: Git (version control, collaboration), Linux (scripting, deployment), VS Code / PyCharm / IntelliJ / Eclipse (IDEs), Docker (containerization, reproducible ML environments).
- **Libraries & Frameworks**: PyTorch & TensorFlow (CNN/RNN, GPU-accelerated DL), JAX (high-performance ML research), pandas / NumPy / Matplotlib / Scikit-learn (data processing, visualization, classical ML).

Honors and Awards

- Finalist - **Madhacks Fall 2023 -** Selected among top teams for innovative software solutions in competitive hackathon environment.
- Finalist - **MadData Fall 2024 -** Recognized for excellence in data analysis and machine learning application.